

Quantum radiation emitted by a rotating nanoparticle

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1 Introduction

A neutral object undergoing a non-uniform motion can emit radiation, a phenomenon known as the dynamical Casimir effect. The purpose of these notes is to show how a rotating nanoparticle in a vacuum can produce such an effect.

The persistence of zero-point fluctuations of the electromagnetic field induces multipoles on polarizable objects, even in the vacuum state. The scattered fields, however, are heavily affected by the object's rotation, producing a modulation in frequency of the field's modes. The mixing of frequencies leads to the generation of photons.

2 Scattering and multipole fields

The total electric field is the sum of the free field and the scattered field. This can be done in two equivalent ways: by propagating the scattered field with its Retarded Green's function, or by propagating with the advanced Green's function. Considering the dipole approximation, the total field can be written as

$$\mathbf{E}(\mathbf{r}, \omega) = \overleftrightarrow{G}^R(\mathbf{r}, \mathbf{r}', \omega) \cdot \mathbf{d}(\omega) + \mathbf{E}^{in}(\mathbf{r}, \omega) = \overleftrightarrow{G}^A(\mathbf{r}, \mathbf{r}', \omega) \cdot \mathbf{d}(\omega) + \mathbf{E}^{out}(\mathbf{r}, \omega), \quad (1)$$

where $\mathbf{r}'$ is the dipole position, $\mathbf{E}^{in}(\mathbf{r}, \omega)$ is the input field, which corresponds to the free field at times $t \rightarrow -\infty$ and $\mathbf{E}^{out}(\mathbf{r}, \omega)$ is the output field, the free field at the opposite limit. The scattering changes the equivalence between those fields, resulting in

$$\mathbf{E}^{out}(\mathbf{r}, \omega) = \overleftrightarrow{\delta G}(\mathbf{r}, \mathbf{r}', \omega) \cdot \mathbf{d}(\omega) + \mathbf{E}^{in}. \quad (2)$$

The function $\overleftrightarrow{\delta G}(\mathbf{r}, \mathbf{r}', \omega)$ is the difference between the retarded and advanced Green's functions, which shall be referred to as demi-advanced Green's function.

To determine the relationship between the output field operators and the input ones, it is necessary to choose a basis to describe the modes of the field. Due to the spherical symmetry of the problem, the most adequate choice is to decompose the field as a sum of multipole waves. Setting the electric dipole at $\mathbf{r}' = \mathbf{0}$, one has the following expressions for the field and the demi-advanced Green's function in the radiation zone

$$\begin{aligned} \mathbf{E}(\mathbf{r}, \omega) &= i\mathcal{E}(\omega) \sum_{JM\lambda} a_{JM\lambda}(\omega) \theta(\omega) \mathbf{I}_{JM\lambda}(\mathbf{r}) - a_{JM\lambda}^\dagger(-\omega) \theta(-\omega) \mathbf{I}_{JM\lambda}^*(\mathbf{r}), \quad \mathcal{E}(\omega) = \sqrt{\frac{\hbar\omega}{2\epsilon_0 c}} \\ \overleftrightarrow{\delta G}(\mathbf{r}, \omega) &= \frac{2i\omega^2 \sin\left(\frac{\omega r}{c}\right)}{4\pi\epsilon_0 c^2 r} \left[\mathbb{1} - \frac{\mathbf{r} \otimes \mathbf{r}}{r^2} \right]. \end{aligned} \quad (3)$$

The indexes J and M are associated with the angular momentum and its z component, respectively. The index λ indicates if the wave is an electric multipole field or a magnetic multipole field. The

functions $I_{JM\lambda}(\mathbf{r})$ are the vector spherical harmonics, which, at the radiation zone, are given by[1]

$$\begin{aligned}\mathbf{I}_{JME}(\mathbf{r}) &= 4\pi(i)^{J-1} \frac{\cos(\frac{\omega r}{c} - J\frac{\pi}{2})}{r} \mathbf{\Psi}_{JM}(\hat{\mathbf{r}}), \quad \mathbf{\Psi}_{JM} = \frac{1}{\sqrt{J(J+1)}} \nabla_{\hat{\mathbf{r}}} Y_{JM} \\ \mathbf{I}_{JMB}(\mathbf{r}) &= 4\pi(i)^J \frac{\sin(\frac{\omega r}{c} - J\frac{\pi}{2})}{r} \mathbf{\Phi}_{JM}(\hat{\mathbf{r}}), \quad \mathbf{\Phi}_{JM} = \frac{1}{\sqrt{J(J+1)}} \hat{\mathbf{r}} \times \nabla_{\hat{\mathbf{r}}} Y_{JM}.\end{aligned}\tag{4}$$

Those definitions allow for an expansion $\overleftrightarrow{\delta G}(\mathbf{r}, \omega) \cdot \mathbf{d}(\omega)$ as a sum of vector spherical harmonics.

Since the retarded Green's function is the propagator for an electric dipole's field, it has no radial components in the radiation zone limit. Neither does the demi-advanced Green's function. Thus, using spherical coordinates

$$\begin{aligned}\hat{\boldsymbol{\theta}} \cdot \overleftrightarrow{\delta G}(\mathbf{r}, \omega) \cdot \mathbf{d}(\omega) &= \frac{2i\omega^2 \sin(\frac{\omega r}{c})}{4\pi\epsilon_0 c^2 r} (\cos\theta \cos\varphi d_x(\omega) + \cos\theta \sin\varphi d_y(\omega) - \sin\theta d_z(\omega)) \\ \hat{\boldsymbol{\varphi}} \cdot \overleftrightarrow{\delta G}(\mathbf{r}, \omega) \cdot \mathbf{d}(\omega) &= \frac{2i\omega^2 \sin(\frac{\omega r}{c})}{4\pi\epsilon_0 c^2 r} (-\sin\varphi d_x(\omega) + \cos\varphi d_y(\omega)),\end{aligned}\tag{5}$$

where $d_i(\omega)$ are cartesian components of the dipole operator. Using a table of spherical harmonics, one obtains

$$\overleftrightarrow{\delta G}(\mathbf{r}, \omega) \cdot \mathbf{d}(\omega) = \frac{2i\omega^2 \sin(\frac{\omega r}{c})}{4\pi\epsilon_0 c^2 r} \left\{ -4\sqrt{\frac{\pi}{3}} \left[\left(\frac{\Psi_{11} - \Psi_{1-1}}{2} \right) d_x(\omega) + \left(\frac{\Psi_{11} + \Psi_{1-1}}{2i} \right) d_y(\omega) \right] + \sqrt{\frac{8\pi}{3}} \Psi_{10} d_z(\omega) \right\},\tag{6}$$

showing that the only contribution for the effect comes from electric multipoles with $J = 1$. Writing in terms of the functions $\mathbf{I}_{JME}(\mathbf{r})$ results in

$$\overleftrightarrow{\delta G}(\mathbf{r}, \omega) \cdot \mathbf{d}(\omega) = \frac{2i\omega^2}{(4\pi)^2 \epsilon_0 c^2} \left\{ -2\sqrt{\frac{\pi}{3}} [\mathbf{I}_{11E}(d_x - id_y) + \mathbf{I}_{1-1E}(-d_x - id_y)] + \sqrt{\frac{8\pi}{3}} \mathbf{I}_{10E} d_z \right\}.\tag{7}$$

This relation allows for the equivalence between the creation and annihilation operators for the output field and the components of the dipole. To obtain this equivalence explicitly, it is important to define the positive and negative parts of the dipole operator: Notice that $d_i(\omega) = d_i^+ + d_i^-$. Moreover, the negative part is the hermitian conjugate of the positive part and vice-versa. Using equation 2 and the definition of the electric field, the equivalence between the field operators and the dipole operator is given by

$$\begin{aligned}a_{10E}^{out}(\omega) &= \frac{2\omega^2}{(4\pi)^2 \epsilon_0 c^2 \mathcal{E}(\omega)} \sqrt{\frac{8\pi}{3}} d_z(\omega) + a_{10E}^{in}(\omega) \\ a_{1-1E}^{out}(\omega) &= \frac{4\omega^2}{(4\pi)^2 \epsilon_0 c^2 \mathcal{E}(\omega)} \sqrt{\frac{\pi}{3}} (d_x(\omega) + id_y(\omega)) + a_{1-1E}^{in}(\omega) \\ a_{11E}^{out}(\omega) &= \frac{-4\omega^2}{(4\pi)^2 \epsilon_0 c^2 \mathcal{E}(\omega)} \sqrt{\frac{\pi}{3}} (d_x(\omega) - id_y(\omega)) + a_{11E}^{in}(\omega).\end{aligned}\tag{8}$$

The number of emitted photons of each mode per unit of time is obtained by calculating

$$N_{JME} = \langle 0_{in} | a_{JME}^{out\dagger}(\omega) a_{JME}^{out}(\omega) | 0_{in} \rangle.\tag{9}$$

Since only the electric multipole contributes, the subscript E will be dropped. Hence,

$$\begin{aligned}
\frac{dN_{10}}{d\omega}(\omega) &= \frac{1}{12\pi^3} \frac{\omega^3}{\hbar c^3 \epsilon_0} \langle 0_{in} | d_z^\dagger(\omega) d_z(\omega) | 0_{in} \rangle \\
\frac{dN_{11}}{d\omega}(\omega) &= \frac{1}{48\pi^3} \frac{\omega^3}{\hbar c^3 \epsilon_0} [\langle 0_{in} | d_x^\dagger(\omega) d_x(\omega) | 0_{in} \rangle + \langle 0_{in} | d_y^\dagger(\omega) d_y(\omega) | 0_{in} \rangle \\
&\quad + i(\langle 0_{in} | d_y^\dagger(\omega) d_x(\omega) | 0_{in} \rangle - (\langle 0_{in} | d_x^\dagger(\omega) d_y(\omega) | 0_{in} \rangle))] \\
\frac{dN_{1-1}}{d\omega}(\omega) &= \frac{1}{48\pi^3} \frac{\omega^3}{\hbar c^3 \epsilon_0} [\langle 0_{in} | d_x^\dagger(\omega) d_x(\omega) | 0_{in} \rangle + \langle 0_{in} | d_y^\dagger(\omega) d_y(\omega) | 0_{in} \rangle \\
&\quad - i(\langle 0_{in} | d_y^\dagger(\omega) d_x(\omega) | 0_{in} \rangle - (\langle 0_{in} | d_x^\dagger(\omega) d_y(\omega) | 0_{in} \rangle))].
\end{aligned} \tag{10}$$

3 Dynamical Casimir effect

To calculate the expected values, one considers that the nanoparticle rotates around an axis perpendicular to its symmetry axis. This results in the generation of sidebands with frequency proportional to the rotation frequency. On those conditions, it can be shown that the electric dipole in terms of the dynamical polarizability is given by

$$\mathbf{d}(\omega) = \overleftrightarrow{\alpha}_0(\omega) \cdot \mathbf{E}(\mathbf{0}, \omega) + \overleftrightarrow{\alpha}_+(\omega) \cdot \mathbf{E}(\mathbf{0}, \omega + 2\Omega) + \overleftrightarrow{\alpha}_-(\omega) \cdot \mathbf{E}(\mathbf{0}, \omega - 2\Omega). \tag{11}$$

Therefore, the correlation functions in equation 10 which are the matrix elements elements of $\overleftrightarrow{C} := \langle 0_{in} | \mathbf{d}^\dagger(\omega) \otimes \mathbf{d}(\omega) | 0_{in} \rangle$ are given by (using Einstein's notation)

$$C_{im} = \langle 0_{in} | \mathbf{d}_i^\dagger(\omega) \mathbf{d}_m(\omega) | 0_{in} \rangle = \sum_{\mu, \mu'} \alpha_{\mu ij}^*(\omega) \alpha_{\mu' mn}(\omega) \langle 0_{in} | E_j^\dagger(\mathbf{0}, \omega + 2\mu\Omega) E_n(\mathbf{0}, \omega + 2\mu'\Omega) | 0_{in} \rangle, \tag{12}$$

with $\mu, \mu' = -1, 0, 1$. The field correlation can be calculated explicitly. To do so, one recalls the expression for the quantized electric field in the plane wave basis [2]

$$\mathbf{E}(\mathbf{r}, t) = \int \frac{d^3\mathbf{k}}{(2\pi)^3} \sum_p i \sqrt{\frac{\hbar\omega}{2\epsilon_0}} [a_{\mathbf{k}p}(\omega) e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)} \hat{\mathbf{e}}_p(\hat{\mathbf{k}}) - a_{\mathbf{k}p}^\dagger(-\omega) e^{-i(\mathbf{k}\cdot\mathbf{r}-\omega t)} \hat{\mathbf{e}}_p^*(\hat{\mathbf{k}})]. \tag{13}$$

Thus

$$\mathbf{E}(\mathbf{0}, \omega) = i \sum_p \sqrt{\frac{\hbar|\omega|}{2\epsilon_0}} \frac{\omega^2}{c^3} [\theta(\omega) \int d\Omega_{\mathbf{k}} a_{\mathbf{k}p}(\omega) \hat{\mathbf{e}}_p - \theta(-\omega) \int d\Omega_{\mathbf{k}} a_{\mathbf{k}p}^\dagger(-\omega) \hat{\mathbf{e}}_p^*]. \tag{14}$$

Combining equations 12 and 14, it is straightforward to compute the components C_{im}

$$\begin{aligned}
C_{im}(\omega) &= \frac{\hbar}{2\epsilon_0 c^6} \sum_{\mu\mu'} \sum_{pp'} \alpha_{\mu ij}^*(\omega) \alpha_{\mu' mn}(\omega) (\omega + 2\mu\Omega)^{5/2} (\omega + 2\mu'\Omega)^{5/2} \\
&\quad \langle 0_{in} | [\theta(\omega + 2\mu\Omega) \int d\Omega_{\mathbf{k}} a_{\mathbf{k}p}^\dagger(\omega + 2\mu\Omega) \hat{\mathbf{e}}_{pj} - \theta(-\omega - 2\mu\Omega) \int d\Omega_{\mathbf{k}} a_{\mathbf{k}p}(-\omega - 2\mu\Omega) \hat{\mathbf{e}}_{pj}^*] \\
&\quad [\theta(\omega + 2\mu'\Omega) \int d\Omega_{\mathbf{k}'} a_{\mathbf{k}'p'}(\omega + 2\mu'\Omega) \hat{\mathbf{e}}_{p'n} - \theta(-\omega - 2\mu'\Omega) \int d\Omega_{\mathbf{k}'} a_{\mathbf{k}'p'}^\dagger(-\omega - 2\mu'\Omega) \hat{\mathbf{e}}_{p'n}^*] | 0_{in} \rangle.
\end{aligned} \tag{15}$$

By the properties of the creation and annihilation operators, the non-zero terms are

$$\begin{aligned}
C_{im}(\omega) &= \frac{\hbar}{2\epsilon_0 c^6} \sum_{\mu\mu'} \sum_{pp'} \alpha_{\mu ij}^*(\omega) \alpha_{\mu' mn}(\omega) (\omega + 2\mu\Omega)^{5/2} (\omega + 2\mu'\Omega)^{5/2} \\
&\quad \times \int d\Omega_{\mathbf{k}} \hat{\mathbf{e}}_{pj} \int d\Omega_{\mathbf{k}'} \hat{\mathbf{e}}_{p'n}^* \theta(-\omega - 2\mu\Omega) \theta(-\omega - 2\mu'\Omega) \langle 0_{in} | a_{\mathbf{k}p}(-\omega - 2\mu\Omega) a_{\mathbf{k}'p'}^\dagger(-\omega - 2\mu'\Omega) | 0_{in} \rangle.
\end{aligned} \tag{16}$$

Since the frequency is positive, the contributing terms are $\mu = -1$ and $\mu' = -1$. Therefore

$$C_{im}(\omega) = \frac{\hbar}{2\epsilon_0 c^6} \sum_{pp'} \alpha_{-ij}^*(\omega) \alpha_{-mn}(\omega) (\omega - 2\Omega)^5 \times \int d\Omega_{\mathbf{k}} \hat{\epsilon}_{pj} \int d\Omega_{\mathbf{k}'} \hat{\epsilon}_{p'n}^* \theta(-\omega + 2\Omega) \langle 0_{in} | a_{\mathbf{k}p}(-\omega + 2\Omega) a_{\mathbf{k}'p'}^\dagger(-\omega + 2\Omega) | 0_{in} \rangle. \quad (17)$$

The commutation relations for the creation and annihilation operators follow the relation

$$\langle 0 | [a_{\mathbf{k}p}(\omega), a_{\mathbf{k}'p'}^\dagger(\omega')] | 0 \rangle = \delta^{(3)}(\mathbf{k} - \mathbf{k}') \delta_{pp'}, \quad (18)$$

with $\delta^{(3)}(\mathbf{k} - \mathbf{k}') = \frac{c^3}{\omega^2 \sin \theta} \delta(\omega - \omega') \delta(\theta - \theta') \delta(\phi - \phi')$. The matrix elements $C_{im}(\omega)$ are thus

$$C_{im}(\omega) = \frac{\hbar}{2\epsilon_0 c^3} \frac{T}{2\pi} |\omega - 2\Omega|^3 \theta(-\omega + 2\Omega) \alpha_{-ij}^*(\omega) \alpha_{-mn}(\omega) \sum_p \int_0^\pi d\theta \sin \theta \int_0^{2\pi} d\phi \hat{\epsilon}_{pj}(\theta, \phi) \hat{\epsilon}_{pn}(\theta, \phi). \quad (19)$$

Recalling the expression of the effective polarizabilities and integrating over the angular variables, one finds

$$\overleftrightarrow{C} = \frac{4}{3} \frac{\hbar T}{\epsilon_0 c^3} \frac{|\Delta(\omega - \Omega)|^2}{4} |\omega - 2\Omega|^3 \theta(2\Omega - \omega) \begin{pmatrix} 1 & i & 0 \\ -i & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (20)$$

where

$$\Delta(\omega) = \frac{\alpha_{\parallel}(\omega) - \alpha_{\perp}(\omega)}{2} \quad (21)$$

is the difference between the eigenvalues of the polarizability tensor describing the particle at rest.

It is possible to estimate the photon emission rate. First, we substitute equation 20, obtaining

$$\begin{aligned} \frac{dN_{10}}{d\omega}(\omega) &= 0 \\ \frac{dN_{11}}{d\omega}(\omega) &= \frac{1}{36\pi^3} \frac{T}{c^6 \epsilon_0^2} |\Delta(\omega - \Omega)|^2 \omega^3 |\omega - 2\Omega|^3 \theta(2\Omega - \omega) \\ \frac{dN_{1-1}}{d\omega} &= 0. \end{aligned} \quad (22)$$

The total photon emission is thus

$$N = \int_0^\infty d\omega \frac{dN_{11}}{d\omega}(\omega). \quad (23)$$

It is worth noticing that the spectra of emitted photons is symmetric around $\omega = \Omega$. This can be seen by proving that $\frac{dN}{d\omega}(\omega) = \frac{dN}{d\omega}(2\Omega - \omega)$:

$$\frac{dN}{d\omega}(2\Omega - \omega) = (2\Omega - \omega)^3 |\omega|^3 |\Delta(\Omega - \omega)|^2 = \omega^3 |\omega - 2\Omega|^3 |\Delta(\omega - \Omega)|^2 = \frac{dN}{d\omega}(\omega). \quad (24)$$

Here, it was used that $0 < \omega < 2\Omega$ and the reality property of $\Delta(\omega)$: $\Delta(\omega) = \Delta^*(-\omega)$.

The number of emitted photons during an interval T is

$$N = \frac{1}{9\pi} \frac{T}{c^6 (4\pi\epsilon_0)^2} \int_0^{2\Omega} d\omega \omega^3 |\omega - 2\Omega|^3 |\Delta(\omega - \Omega)|^2 \quad (25)$$

Since the polarizability depends on the electromagnetic behavior of the rotating material, further analysis is necessary to determine this dynamic Casimir effect on different materials.

4 Examples

4.1 Perfectly-Reflecting Nanodumbbell

For a perfect reflector, the polarizability does not depend on the frequencies of the incident waves. This simplifies equation (25) to

$$N = \frac{32}{315\pi} \frac{T}{c^6 (4\pi\epsilon_0)^2} \Delta^2 \Omega^7. \quad (26)$$

From reference [3] one can determine the difference between the longitudinal and transverse polarizabilities for a conducting dumbbell

$$\Delta = 10\pi\epsilon_0\zeta(3)R^3, \quad (27)$$

where R is the radius of each sphere of the dumbbell. Hence, the photon emission rate Γ (number of photons per unit of time) is

$$\Gamma = \frac{40\zeta^2(3)}{63\pi} \left(\frac{\Omega R}{c} \right)^6 \Omega \quad (28)$$

Following the experiment presented in reference [4], the experimental values for R and Ω are $\Omega = 2\pi \times 5.2$ GHz and $R = 150$ nm (R = diameter of each nanosphere in the nanodumbbell). Under those conditions, the emission rate is $\Gamma \sim 6 \times 10^{-19} \text{ s}^{-1}$, indicating a very small effect.

Note that experimental parameters are such that $\Omega R = 4.9 \times 10^3$ m/s, which is close to the speed of sound in amorphous silica $v_{\text{sound}} \sim 6.0 \times 10^3$ m/s [5]. Now, we take new values of frequency and radius such that $\Omega' R' = 4.9 \times 10^3$ m/s as before and the smallest possible value of R' still compatible with our approach based on the macroscopic Maxwell equations: $R' = 1$ nm. In this case, we find $\Omega' = 2\pi \times 0.78$ THz and $\Gamma' = 150 \Gamma = 10^{-16} \text{ s}^{-1}$ which is still very small.

4.2 Perfectly-reflecting spheroid

$$\Delta = \frac{\alpha_{\parallel} - \alpha_{\perp}}{2} = 4\pi\epsilon_0 \frac{r_{\parallel} r_{\perp}^2}{6} \frac{(N_{\perp} - N_{\parallel})}{N_{\perp} N_{\parallel}}. \quad (29)$$

where $r_{\parallel}$ and $r_{\perp}$ are the longitudinal and transverse radius of the spheroid, respectively. The dimensionless constants $N_{\perp}$ and $N_{\parallel}$ called depolarizing factors. Their expression in terms of the dimensions of a spheroid is

$$N_i = \frac{r_{\parallel} r_{\perp}^2}{2} \int_0^{\infty} \frac{ds}{(s + r_i^2)(s + r_{\perp}^2) \sqrt{s + r_{\parallel}^2}}. \quad (30)$$

Those constants have an expression in terms of elementary functions [6]

$$\begin{aligned} N_{\parallel} &= \frac{1 - e^2}{2e^3} \left(\log \frac{1 + e}{1 - e} - 2e \right) \\ N_{\perp} &= \frac{1}{2}(1 - N_{\parallel}) \\ e &= \sqrt{1 - \frac{r_{\perp}^2}{r_{\parallel}^2}}. \end{aligned} \quad (31)$$

For example, one can calculate the depolarizing factors for $r_{\perp}/r_{\parallel} = 1/2$, finding $N_{\parallel} = 0.17$ and $N_{\perp} = 0.415$. The photon emission rate in those conditions (which shall be referenced by Γ_0 henceforth) is

$$\Gamma_0 = \frac{8}{2835\pi} \left[\frac{r_{\parallel} r_{\perp}^2}{c^3} \left(\frac{1}{N_{\parallel}} - \frac{1}{N_{\perp}} \right) \right]^2 \Omega^7. \quad (32)$$

4.3 Dielectric spheroid

To increase the effect, it is possible to explore the dispersive properties of the materials. For a spheroid, the (resting) longitudinal and transverse polarizabilities for a material whose permittivity ϵ depends on the frequency are given by [6, 7]

$$\begin{aligned}\alpha_{\parallel}(\omega) &= 4\pi\epsilon_0 \frac{r_{\parallel}r_{\perp}^2}{3} \frac{\epsilon(\omega) - 1}{1 + N_{\parallel}(\epsilon(\omega) - 1)} \\ \alpha_{\perp}(\omega) &= 4\pi\epsilon_0 \frac{r_{\parallel}r_{\perp}^2}{3} \frac{\epsilon(\omega) - 1}{1 + N_{\perp}(\epsilon(\omega) - 1)},\end{aligned}\tag{33}$$

For a dielectric material, the permittivity can be obtained by the Lorentz model [8]:

$$\epsilon(\omega) = \epsilon_{\infty} \left(1 + \frac{\omega_L^2 - \omega_T^2}{\omega_T^2 - \omega^2 - i\omega\gamma} \right),\tag{34}$$

where $\epsilon_{\infty} = 2.9$, $\omega_L = 1.3 \times 10^{10}$ Hz, $\omega_T = 5.7 \times 10^9$ Hz and $\gamma = 2.8 \times 10^8$ Hz are optical parameters. First, let us write the difference between longitudinal and transverse polarizabilities since it is this difference that appears on the photon emission rate

$$\Delta(\omega) = \frac{\alpha_{\parallel} - \alpha_{\perp}}{2} = 4\pi\epsilon_0 \frac{r_{\parallel}r_{\perp}^2}{6} \frac{(N_{\perp} - N_{\parallel})(\epsilon(\omega) - 1)^2}{N_{\perp}N_{\parallel}(\epsilon(\omega) - 1 + 1/N_{\parallel})(\epsilon(\omega) - 1 + 1/N_{\perp})}.\tag{35}$$

Thus, considering $\omega_L, \omega_T \gg \gamma$, one finds the two resonating frequencies to be $\omega_i = \sqrt{\omega_T^2 - \frac{\epsilon_{\infty}(\omega_L^2 - \omega_T^2)}{1 - 1/N_i - \epsilon_{\infty}}}$. For a BST spheroid of radii 1 nm and 2 nm those values are 1.1×10^{10} rad/s and 0.9×10^{10} rad/s (see figure 14). However, most materials are modeled as a collection of oscillators, and thus the permittivity is written as a sum of the contributions of each oscillator [9]

$$\epsilon(\omega) = 1 + \sum_n \frac{C_n}{1 - \left(\frac{\omega}{\omega_n}\right)^2 - i\frac{\omega\gamma_n}{\omega_n^2}},\tag{36}$$

where C_n is the strength of oscillator n and ω_n its corresponding resonating frequency. That is the case for the silica (SiO_2), which the Lorentz model parameters can be found on reference [9]. Another possibility is the barium strontium titanate (BST), a material whose resonating frequencies are of the order of GHz [8].

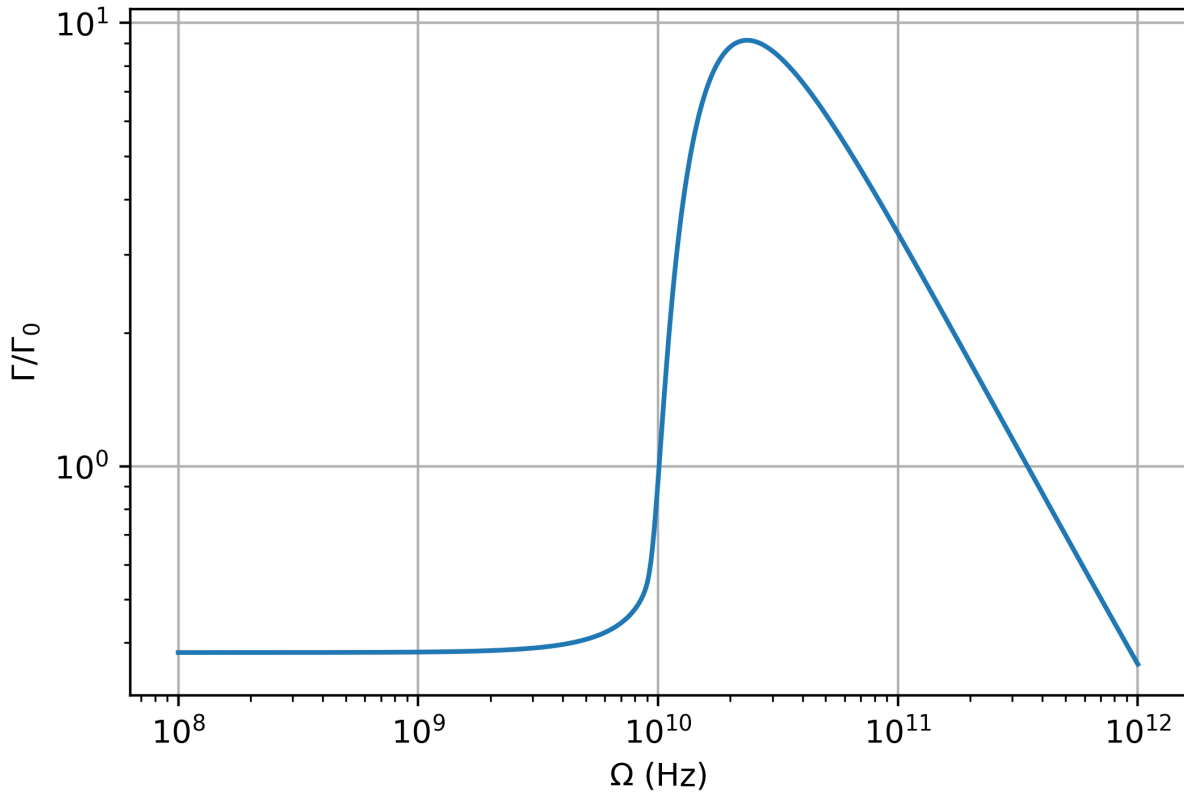
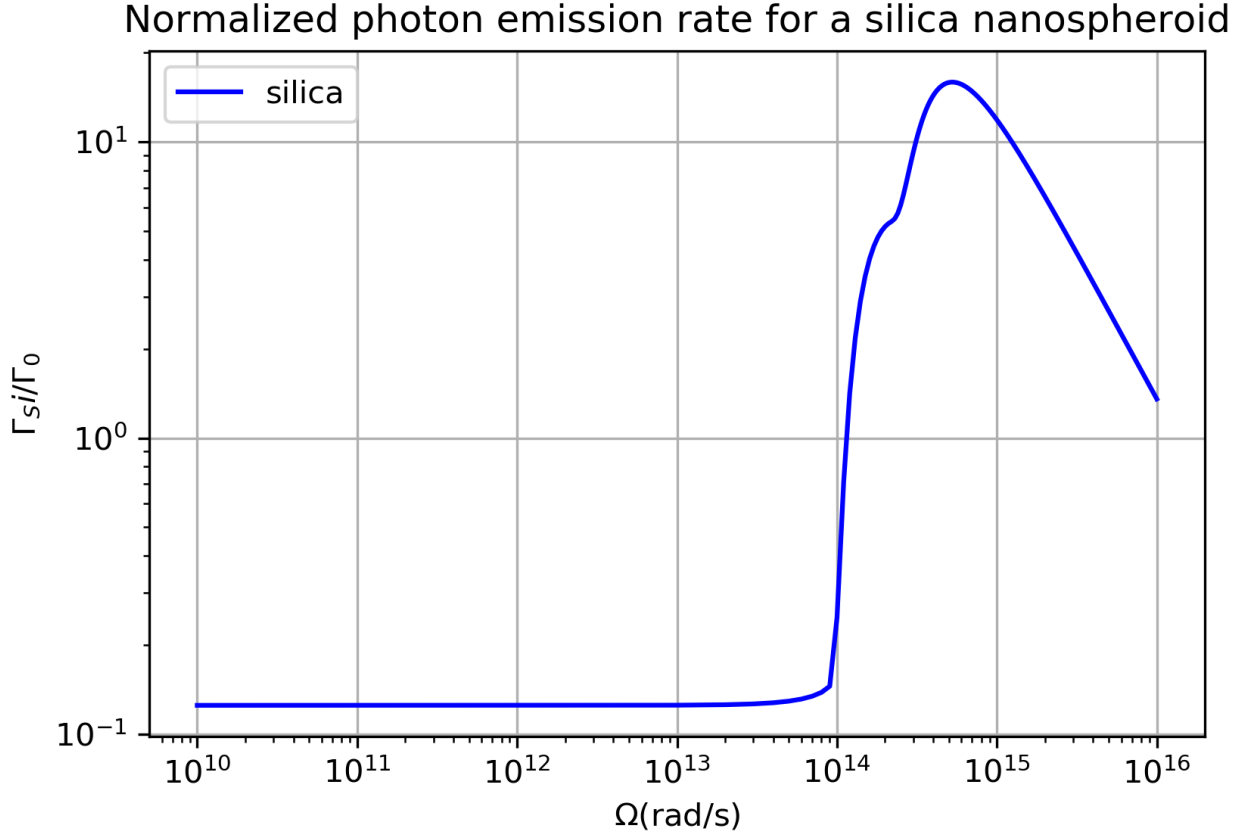


Figure 1: Photon emission rates normalized by the perfect reflector emission (Γ_0 in eq. 32) for spheroids of different materials: silica and BST. The radii of the spheroids are considered 1 nm and 2 nm.

As a consistency check, it is possible to compare our numerical results with the analytical result for the photon emission in the quasi-static regime Γ_S ($\Delta(\omega) \approx \Delta(0)$),

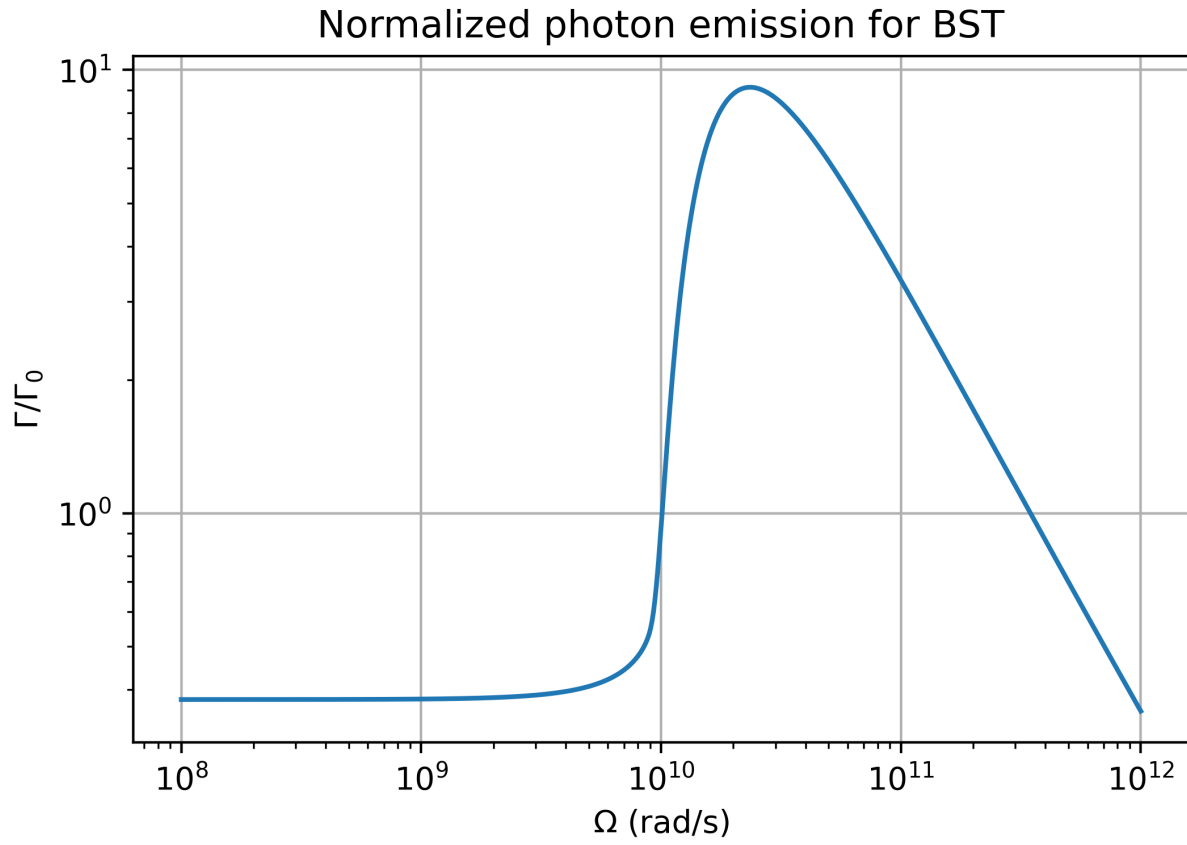
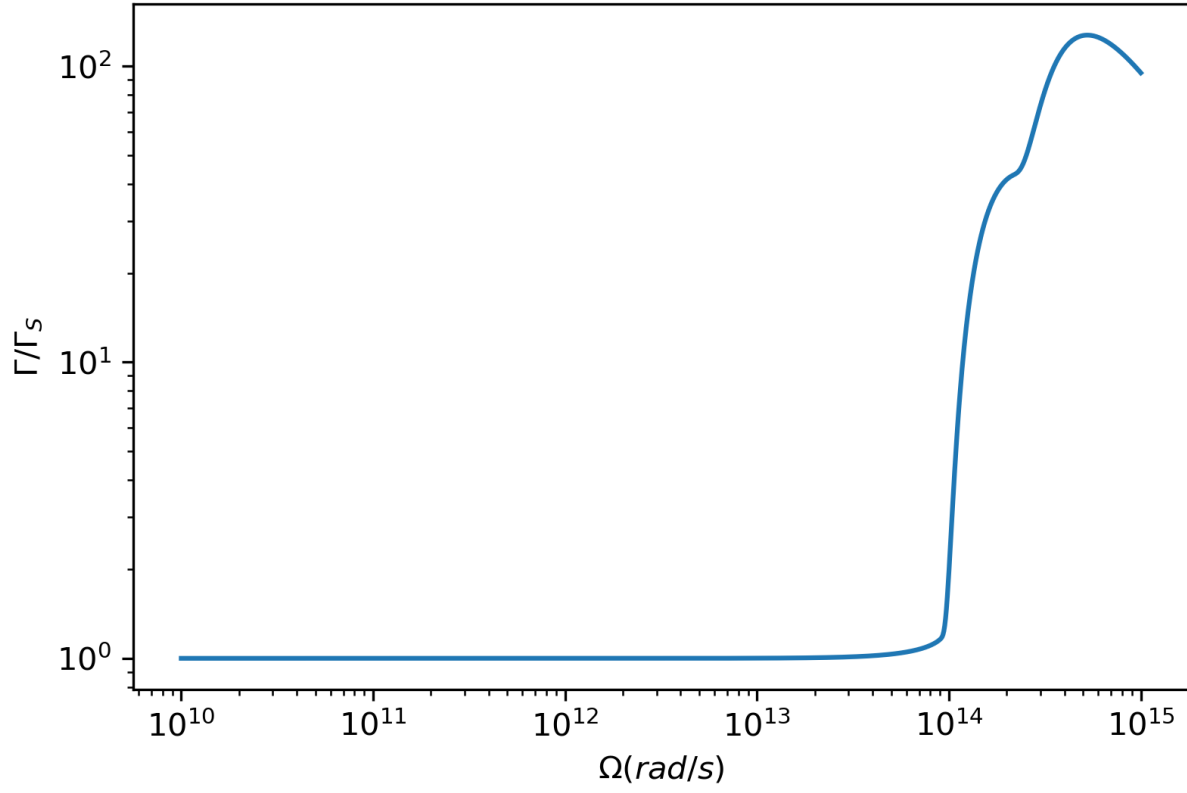


Figure 2: Photon emission rates normalized by the quasi-static regime for spheroids of different materials: silica and BST. The radii of the spheroids are considered 1 nm and 2 nm.

4.4 Metals

If the material that composes the rotating object is a metal, the most suitable model for the electric permittivity is the Plasma-Drude model. Let us start without the dissipation term (plasma)

$$\epsilon(\omega) = 1 - \frac{\omega_p^2}{\omega^2} \quad (37)$$

By modeling a metal as plasma (equation 37), it is worth establishing its domain of validity. To determine this, one can calculate explicitly the photon emission rate using equations 35 and 25

$$\Gamma = \frac{1}{9\pi c^6} \frac{(r_{\parallel} r_{\perp}^2)^2}{36} \int_0^{2\Omega} d\omega \omega^3 |\omega - 2\Omega|^3 \left(\frac{1}{N_{\parallel}} - \frac{1}{N_{\perp}} \right)^2 \left| \frac{\left(\frac{\omega_p}{\omega - \Omega} \right)^4}{\left[-\left(\frac{\omega_p}{\omega - \Omega} \right)^2 + 1/N_{\parallel} \right] \left[-\left(\frac{\omega_p}{\omega - \Omega} \right)^2 + 1/N_{\perp} \right]} \right|^2. \quad (38)$$

This integral can become dimensionless by introducing the new variable $u = \frac{\Omega - \omega}{\Omega}$. Let us focus on the integral and write

$$\Gamma = \frac{1}{9\pi} \frac{\Omega^7}{c^6} \frac{(r_{\parallel} r_{\perp}^2)^2}{36} \left(\frac{1}{N_{\parallel}} - \frac{1}{N_{\perp}} \right)^2 \int_{-1}^1 du (1 - u^2)^3 \left[\frac{\left(\frac{\omega_p^2}{u^2 \Omega^2} \right)^2}{\left(\frac{1}{N_{\parallel}} - \frac{\omega_p^2}{u^2 \Omega^2} \right) \left(\frac{1}{N_{\perp}} - \frac{\omega_p^2}{u^2 \Omega^2} \right)} \right]^2, \quad (39)$$

leading to

$$\Gamma = \frac{1}{9\pi} \frac{\Omega^7}{c^6} \frac{(r_{\parallel} r_{\perp}^2)^2}{36} \left(\frac{1}{N_{\parallel}} - \frac{1}{N_{\perp}} \right)^2 \int_{-1}^1 du (1 - u^2)^3 \frac{1}{\left(\frac{\Omega^2 u^2}{\omega_p^2 N_{\perp}} - 1 \right)^2 \left(\frac{\Omega^2 u^2}{\omega_p^2 N_{\parallel}} - 1 \right)^2}. \quad (40)$$

For the integral to converge, it is necessary that $\frac{\Omega^2 u^2}{\omega_p^2 N_i} - 1 \neq 0$, leaving the condition $u \neq \frac{\omega_p \sqrt{N_i}}{\Omega}$. In addition, for the singularity to be outside the integration region, one must have $\frac{\omega_p \sqrt{N_i}}{\Omega} > 1$, resulting in $\omega_p > \frac{\Omega}{\sqrt{N_i}}$. Thus the importance of dissipation, as the poles are shifted from the real axis to other regions of the complex plane and allowing convergence for all frequencies.

Let us consider now a permittivity for a metal considering a dissipation term:

$$\epsilon(\omega) = 1 - \frac{\omega_p^2}{\omega(\omega + i\gamma)} \quad (41)$$

The equation 40 can be generalized using equation 41 in terms of the variable u defined above, resulting in

$$\Gamma = \frac{1}{9\pi} \frac{\Omega^7}{c^6} \frac{(r_{\parallel} r_{\perp}^2)^2}{36} \left(\frac{1}{N_{\parallel}} - \frac{1}{N_{\perp}} \right)^2 \int_{-1}^1 du (1 - u^2)^3 \frac{1}{\left[\left(1 - \frac{\Omega^2 u^2}{\omega_p^2 N_{\perp}} \right)^2 + \left(\frac{\Omega \gamma u}{\omega_p^2 N_{\perp}} \right)^2 \right] \left[\left(1 - \frac{\Omega^2 u^2}{\omega_p^2 N_{\parallel}} \right)^2 + \left(\frac{\Omega \gamma u}{\omega_p^2 N_{\parallel}} \right)^2 \right]}. \quad (42)$$

Notice that if $\gamma \rightarrow 0$, one obtains the result for a plasma (equation 40).

There are 3 important limits for the above expression. The first is the quasi-static regime, on which $\omega_p \gg \Omega$. Given this approximation, the emission peaks lie outside the integration regions, leading to a constant denominator. Hence, one obtains the result from equation 32. The other limits are the resonant and high-frequency regimes.

Since the emission spectra is given by four sharp peaks with a width determined by γ (which is typically much smaller than ω_p), the points contributing to the integral are those close to maximum values, given by $u_i = \omega_p \sqrt{N_i} / \Omega$. Hence, the functions are constant on those values, except for terms

involving $(1 - \Omega u / \omega_p N_i)$. The complete expression becomes a sum of two peaks, and the integration limits become infinite. **Due to the symmetry of the spectra, the limits of integration can be changed from 0 to ∞**

$$\Gamma = \frac{1}{9\pi} \frac{\Omega^7}{c^6} \frac{(r_{\parallel} r_{\perp}^2)^2}{36} \left(\frac{1}{N_{\parallel}} - \frac{1}{N_{\perp}} \right)^2 \times 2 \int_{-\infty}^{\infty} du \left\{ \frac{(1 - u_{\perp}^2)^3}{\left[4 \left(1 - \frac{u}{u_{\perp}} \right)^2 + \frac{\gamma^2}{\omega_p^2 N_{\perp}} \right] \left[\left(1 - \frac{u_{\perp}^2}{u_{\parallel}^2} \right)^2 + \frac{\gamma^2 N_{\perp}}{\omega_p^2 N_{\parallel}^2} \right]} + \frac{(1 - u_{\parallel}^2)^3}{\left[4 \left(1 - \frac{u}{u_{\parallel}} \right)^2 + \frac{\gamma^2}{\omega_p^2 N_{\parallel}} \right] \left[\left(1 - \frac{u_{\parallel}^2}{u_{\perp}^2} \right)^2 + \frac{\gamma^2 N_{\parallel}}{\omega_p^2 N_{\perp}^2} \right]} \right\}. \quad (43)$$

The integral can be calculated explicitly. The result is given by

$$\Gamma = \frac{1}{9} \frac{\Omega^6 \omega_p^2}{\gamma c^6} \frac{(r_{\parallel} r_{\perp}^2)^2}{36} \left(\frac{1}{N_{\parallel}} - \frac{1}{N_{\perp}} \right)^2 \left\{ N_{\perp} \frac{(1 - u_{\perp}^2)^3}{\left(1 - \frac{u_{\perp}^2}{u_{\parallel}^2} \right)^2 + \frac{\gamma^2 N_{\perp}}{\omega_p^2 N_{\parallel}^2}} + N_{\parallel} \frac{(1 - u_{\parallel}^2)^3}{\left(1 - \frac{u_{\parallel}^2}{u_{\perp}^2} \right)^2 + \frac{\gamma^2 N_{\parallel}}{\omega_p^2 N_{\perp}^2}} \right\} \quad (44)$$

This result takes into account all regimes, especially if $\Omega \sim \omega_p$. The high frequency limit, on which $\Omega \gg \omega_p$ reduces the above expression to

$$\Gamma = \frac{1}{9} \frac{\Omega^6 \omega_p^2}{\gamma c^6} \frac{(r_{\parallel} r_{\perp}^2)^2}{36} \left(\frac{1}{N_{\parallel}} - \frac{1}{N_{\perp}} \right)^2 \left\{ \frac{N_{\perp}}{\left(1 - \frac{u_{\perp}^2}{u_{\parallel}^2} \right)^2 + \frac{\gamma^2 N_{\perp}}{\omega_p^2 N_{\parallel}^2}} + \frac{N_{\parallel}}{\left(1 - \frac{u_{\parallel}^2}{u_{\perp}^2} \right)^2 + \frac{\gamma^2 N_{\parallel}}{\omega_p^2 N_{\perp}^2}} \right\}. \quad (45)$$

Another regime worth discussing is when $N_{\perp} \approx N_{\parallel}$, which shall be referred to as the quasi-degenerate regime. Within this condition, the spectra become a squared lorentzian distribution, instead of a sum of two regular lorentzian peaks as before. The expression 43 is thus transformed to

$$\Gamma = \frac{1}{9\pi} \frac{\Omega^7}{c^6} \frac{(r_{\parallel} r_{\perp}^2)^2}{36} \left(\frac{1}{N_{\parallel}} - \frac{1}{N_{\perp}} \right)^2 \int_{-\infty}^{\infty} du \frac{(1 - u_{\perp}^2)^3}{\left[4 \left(1 - \frac{u}{u_{\perp}} \right)^2 + \frac{\gamma^2}{\omega_p^2 N_{\perp}} \right]^2}. \quad (46)$$

Integrating analytically results in

$$\Gamma = \frac{1}{36} \frac{\Omega^6 \omega_p^4}{\gamma^3 c^6} \frac{(r_{\parallel} r_{\perp}^2)^2}{36} \left(\frac{1}{N_{\parallel}} - \frac{1}{N_{\perp}} \right)^2 N_{\perp}^2 (1 - u_{\perp}^2)^3. \quad (47)$$

Parameters used for the following plots were: $N_{\perp} = 0.15$, $N_{\parallel} = 0.7$, $r_{\parallel} = 2r_{\perp} = 2 \cdot 10^{-9}$ m, $\omega_p = 10^{10}$ rad/s

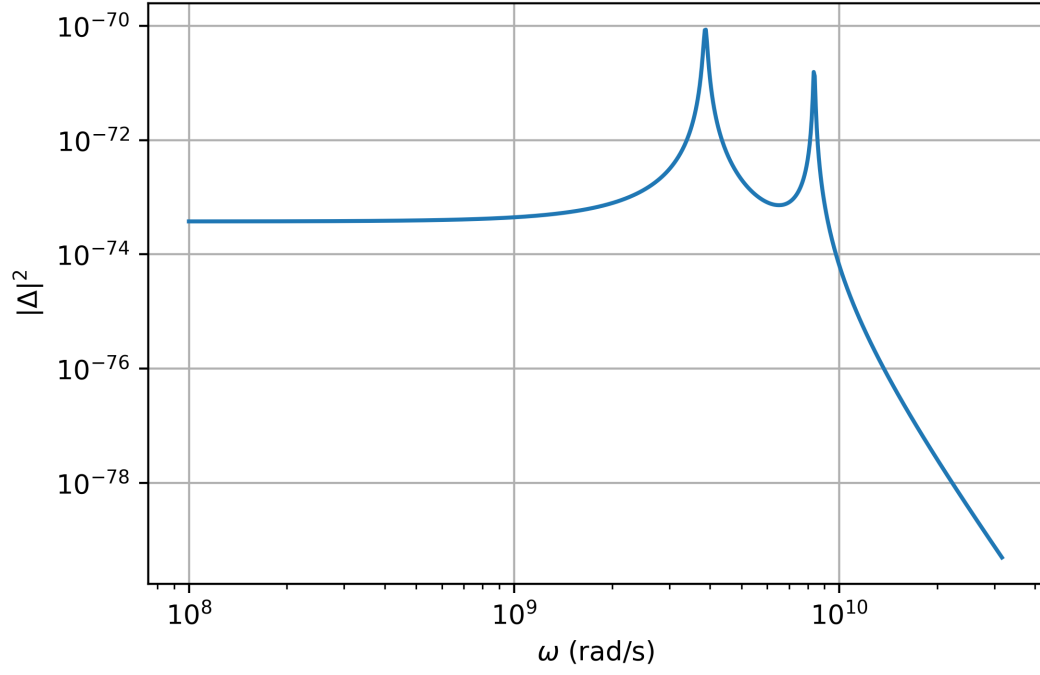


Figure 3: Squared modulus of the difference between transverse and longitudinal polarizabilities. Peaks of $|\Delta|^2$: $3.88 \cdot 10^9$ rad/s and $8.34 \cdot 10^9$ rad/s. Analytically, peaks should be at $\omega = \sqrt{\omega_p^2 N_{\parallel, \perp}}$, with numerical values: $3.87 \cdot 10^9$ rad/s and $8.36 \cdot 10^9$ rad/s.

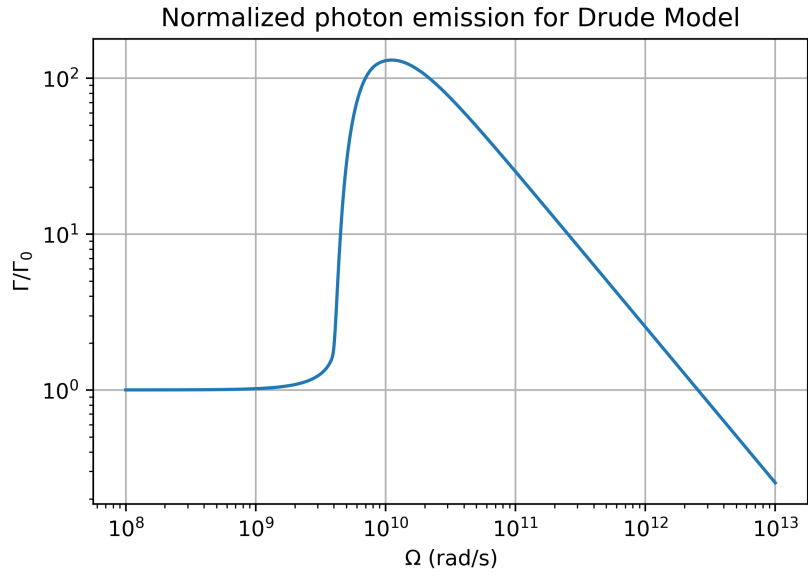


Figure 4: Photon emission for gold [10], $\gamma = 4 \cdot 10^{-3} \omega_p$.

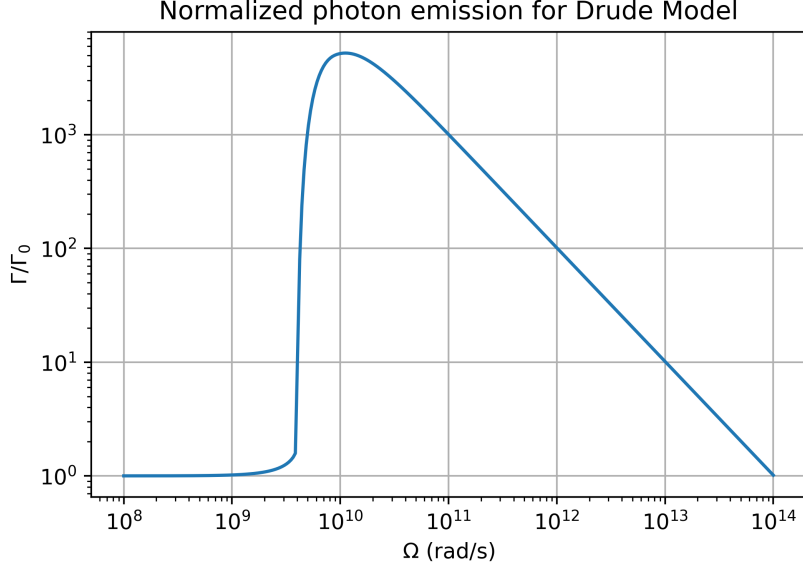


Figure 5: Photon emission rate for a Drude metal considering $\omega_p = 10^{10}\text{rad/s}$ and $\gamma = 10^{-4}\omega_p$

4.5 Dependence with size and format

All the calculations to this point consider a nanoparticle with a fixed radius. In this section, we aim to optimize the effect given the sound speed limit, above which the nanoparticle can be damaged. Therefore, the angular frequency shall be substituted by

$$\Omega = \frac{v_s}{r_{\parallel}}. \quad (48)$$

Define x as the ratio between the radii of the spheroid, i.e. $x = r_{\perp}/r_{\parallel}$. Supposing the rotating nanoparticle is made of metal, one can apply equation 42. See figure 6 for results.

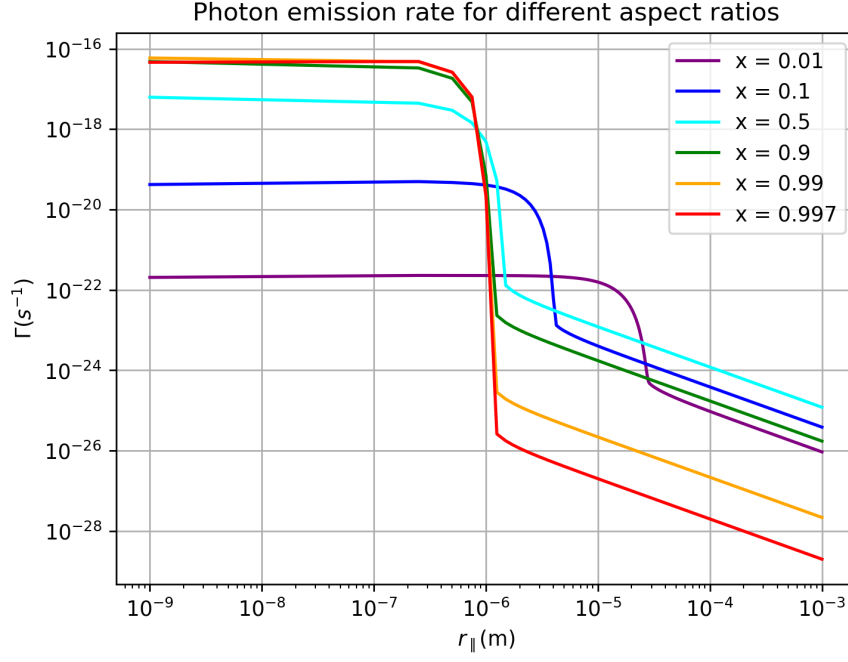


Figure 6: Photon emission rate as a function of the bigger radius of the prolate spheroid for a metal with $\omega_p = 10^{10}$ rad/s. The rotation frequency Ω is such that the velocity of the tip of the spheroid matches the speed of sound: $\Omega r_{||} = v_s = 6 \times 10^3$ m/s.

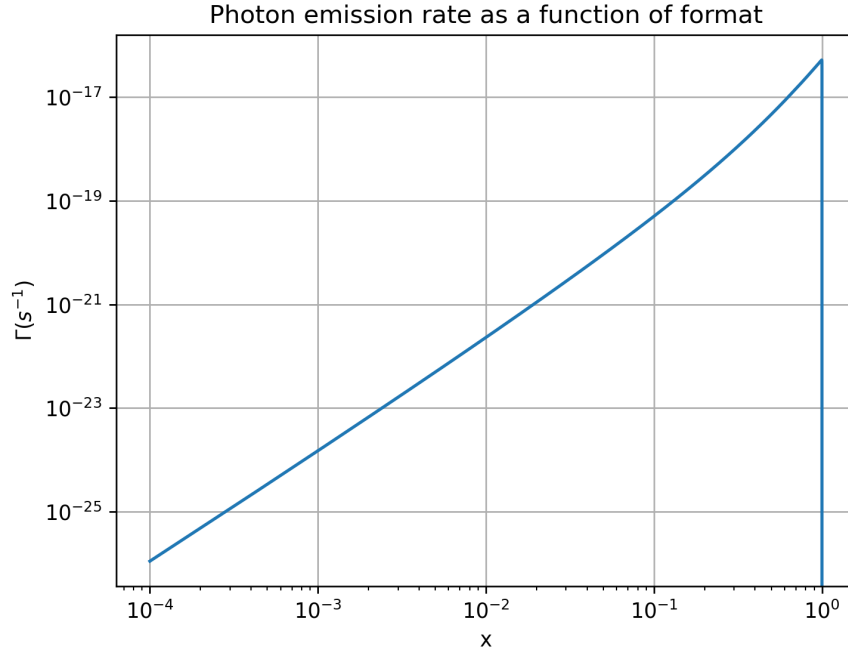


Figure 7: Photon emission rate as a function of $x = r_{\perp}/r_{||}$ for a metallic nanospheroid with radius $r_{||} = 0.2 \mu\text{m}$

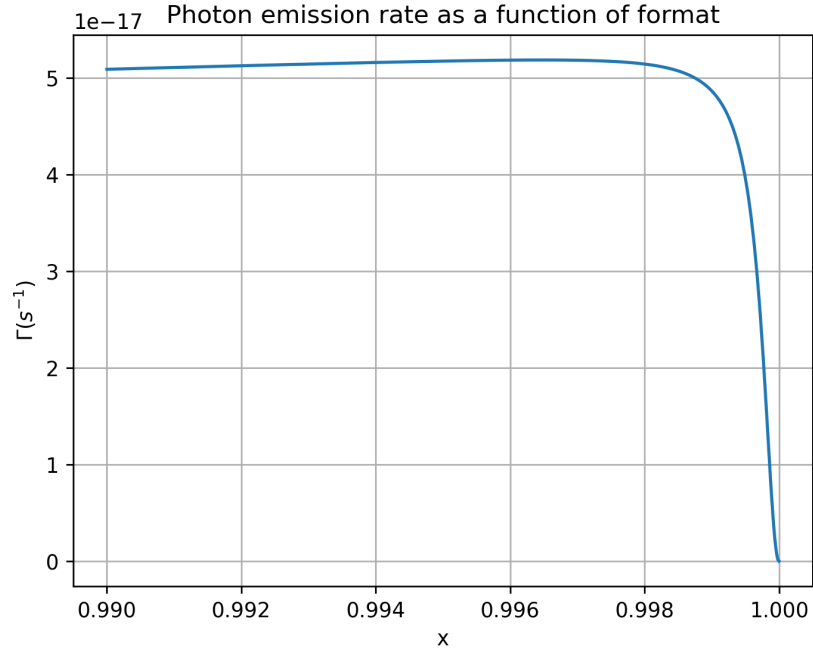


Figure 8: Photon emission rate as a function of $x = r_{\perp}/r_{\parallel}$ for a metallic nanospheroid with radius $r_{\parallel} = 0.2 \mu\text{m}$

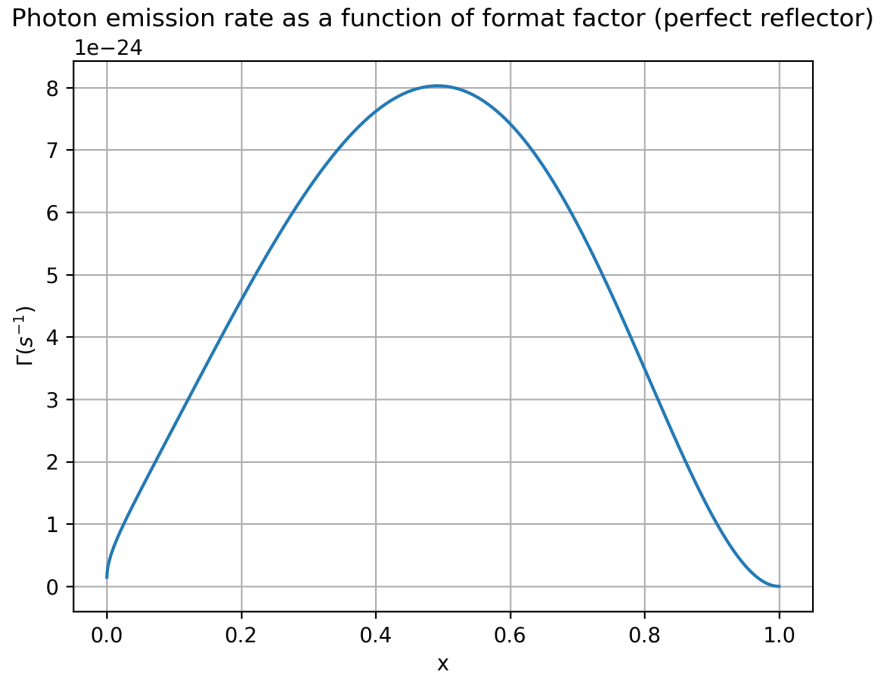


Figure 9: Photon emission rate as a function of $x = r_{\perp}/r_{\parallel}$ for a metallic nanospheroid with radius $r_{\parallel} = 30 \mu\text{m}$

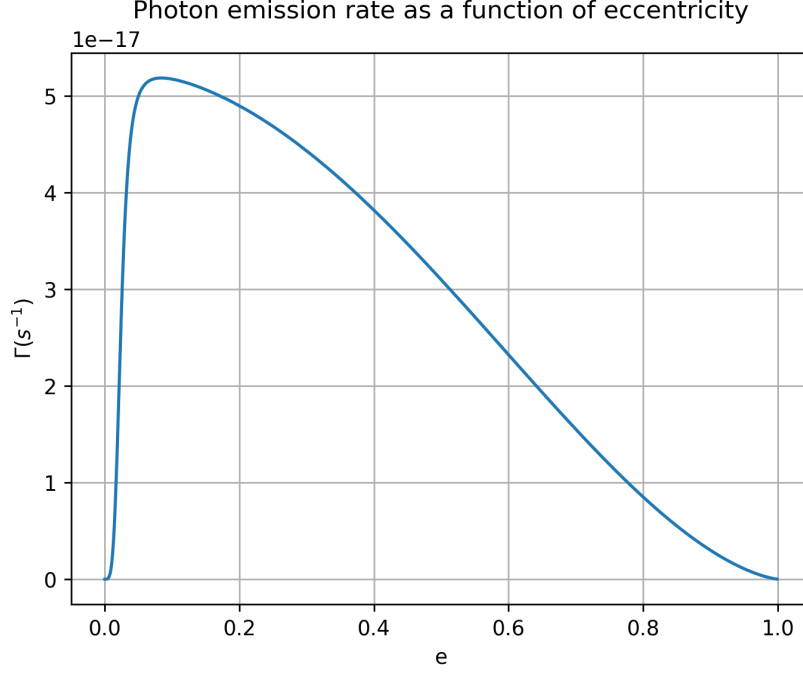


Figure 10: Photon emission rate as a function of the eccentricity $e = \sqrt{1 - x^2}$ $x = r_{\perp}/r_{\parallel}$ for a metallic nanospheroid with radius $r_{\parallel} = 0.2 \mu\text{m}$

5 Dynamical Casimir Effect x Quantum Friction

The quantum friction corresponds to a dissipative force acting on an object due to quantum vacuum fluctuations. For a spheroid rotating around its symmetry axis, the radiated at zero temperature is given by [11]

$$P_{rad} = -\frac{4\hbar}{3\pi c^3} \int_0^{\Omega} d\omega \omega^4 \text{Im}\{\alpha_{\perp}(\omega - \Omega)\}. \quad (49)$$

One can compare the radiated power from equation 49 with the one obtained from the dynamical Casimir effect. This can be done by multiplying the integrand in equation 26. The values for a silica nanospheroid are registered in the graph below.

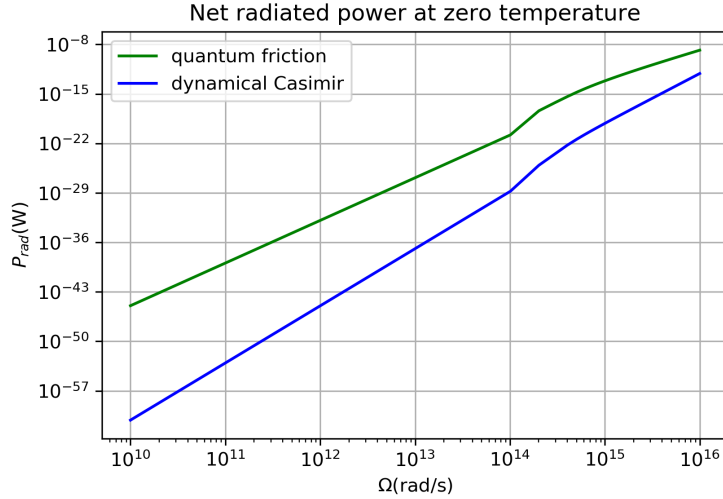


Figure 11: radiated power emitted from the dynamical Casimir effect and from the quantum friction by a silica nanospheroid

As the radiated power depends on the imaginary part of the transverse polarizability, the less dissipation a material has, the smaller the effect of quantum friction.

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Appendix A Polarizability of a rotating object

On reference [11] it is derived the polarizability for an object rotating around its symmetry axis. However, recent experiments [4] achieved the highest rotation frequencies by rotating an object (similar to the figure 12) around an axis perpendicular to its symmetry axis, thus motivating the study of the behavior of the polarizability in this case, which is not yet discussed in the literature. This will be done using the method from Manjavacas&de Abajo [11].

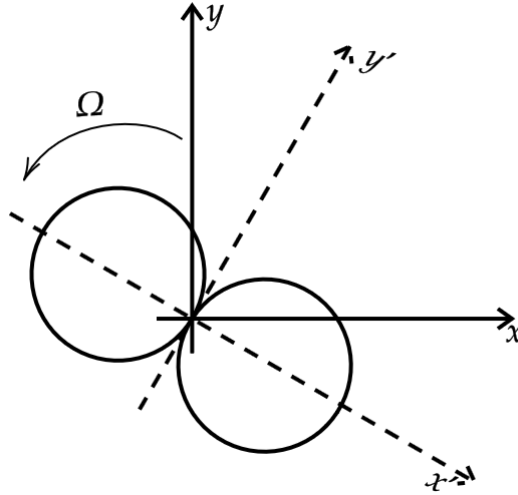


Figure 12: Change of frames between object's comoving frame ($x'y'$) and laboratory frame (xy).

Supposing x to be the symmetry e and the rotation around the z axis with angular velocity Ω , the polarizability on the resting frame is written as

$$\overleftrightarrow{\alpha}(\omega) = \begin{pmatrix} \alpha_{\parallel} & 0 & 0 \\ 0 & \alpha_{\perp} & 0 \\ 0 & 0 & \alpha_{\perp} \end{pmatrix}. \quad (50)$$

Therefore

$$\begin{aligned} d'_x &= \alpha_{\parallel} E'_x \\ d'_y &= \alpha_{\perp} E'_y \\ d'_z &= \alpha_{\perp} E'_z. \end{aligned}$$

Defining the following quantities

$$\begin{aligned} S &= \frac{\alpha_{\parallel}(\omega) + \alpha_{\perp}(\omega)}{2} \\ \Delta &= \frac{\alpha_{\parallel}(\omega) - \alpha_{\perp}(\omega)}{2}. \end{aligned} \quad (51)$$

one obtains

$$\begin{aligned}
d_x(\omega) &= \frac{1}{2}[S^+(E_x(\omega) + iE_y(\omega)) + S^-(E_x(\omega) - iE_y(\omega)) + \\
&\quad + \Delta^+(E'_x(\omega_+) - iE'_y(\omega_+)) + \Delta^-(E'_x(\omega_-) + iE'_y(\omega_-))] \\
d_y(\omega) &= \frac{1}{2}[S^+(-iE_x(\omega) + E_y(\omega)) + S^-(-iE_x(\omega) - E_y(\omega)) - \\
&\quad - \Delta^+(iE'_x(\omega_+) + E'_y(\omega_+)) + \Delta^-(iE'_x(\omega_-) - E'_y(\omega_-))]
\end{aligned} \tag{52}$$

by using a matrix to represent the rotation by the z axis. Superscripts $+$ and $-$ indicates that S e Δ são calculados em ω_+ e ω_- , respectivamente. Embora no caso apresentado na seção anterior o momento de dipolo para uma dada frequência ω é gerado por um campo de mesma frequência, aqui há a geração de bandas laterais, i.e., a frequência do campo é modulada pela frequência de rotação do objeto, de modo que $\mathbf{d}(\omega)$ é gerado por outra componente de Fourier do campo. Esse fenômeno pode ser vinculado ao efeito Casimir dinâmico [?], uma vez que, diferentemente do caso anterior, o objeto não ocupa a mesma posição no espaço para todos os instantes.

A equação 52 envolve ainda componentes do campo no referencial co-móvel à esfera. Isso pode ser resolvido utilizando de forma iterativa as relações dadas pela equação ??, uma vez que são válidas para todo ω

$$\begin{aligned}
E_x(\omega_{--}) + iE_y(\omega_{--}) &= E'_x(\omega_-) + iE'_y(\omega_-) \\
E_x(\omega_{++}) - iE_y(\omega_{++}) &= E'_x(\omega_+) - iE'_y(\omega_+).
\end{aligned} \tag{53}$$

Nesse sentido a equação ?? pode ser entendida como operadores de levantamento e abaixamento da frequência do campo elétrico, em analogia aos operadores escada do oscilador harmônico quântico.

Nessas condições o momento de dipolo é dado pela seguinte equação:

$$\mathbf{d}(\omega) = \overleftrightarrow{\alpha}_0(\omega)\mathbf{E}(\omega) + \overleftrightarrow{\alpha}_+(\omega)\mathbf{E}(\omega + 2\Omega) + \overleftrightarrow{\alpha}_-(\omega)\mathbf{E}(\omega - 2\Omega), \tag{54}$$

na qual $\overleftrightarrow{\alpha}_0(\omega)$, $\overleftrightarrow{\alpha}_+(\omega)$ e $\overleftrightarrow{\alpha}_-(\omega)$ são definidos por

$$\begin{aligned}
\overleftrightarrow{\alpha}_0(\omega) &:= \begin{pmatrix} \frac{1}{2}(S^+ + S^-) & \frac{i}{2}(S^+ - S^-) & 0 \\ \frac{-i}{2}(S^+ - S^-) & \frac{1}{2}(S^+ + S^-) & 0 \\ 0 & 0 & \alpha_\perp(\omega) \end{pmatrix} \\
\overleftrightarrow{\alpha}_+(\omega) &:= \begin{pmatrix} \frac{\Delta^+}{2} & \frac{-i\Delta^+}{2} & 0 \\ \frac{-i\Delta^+}{2} & \frac{-\Delta^+}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
\overleftrightarrow{\alpha}_-(\omega) &:= \begin{pmatrix} \frac{\Delta^-}{2} & \frac{i\Delta^-}{2} & 0 \\ \frac{i\Delta^-}{2} & \frac{-\Delta^-}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}.
\end{aligned} \tag{55}$$

Ressalta-se ao leitor duas peculiaridades do tensor calculado: diferentemente da polarizabilidade calculada na equação ?? este caso pode ocorrer mesmo se o material não for dispersivo, contanto que o material seja anisotrópico de modo a ter $\alpha_\perp \neq \alpha_\parallel$. A segunda propriedade são as bandas laterais moduladas pelo dobro da frequência de rotação, o que pode ser entendido pela simetria de translação temporal discreta do problema, uma vez que o objeto ocupa a mesma posição no espaço a cada metade de período de rotação.

Appendix B Optical properties for the BST

The optical data regarding the dielectric properties of the BST is registered in the graphs below.

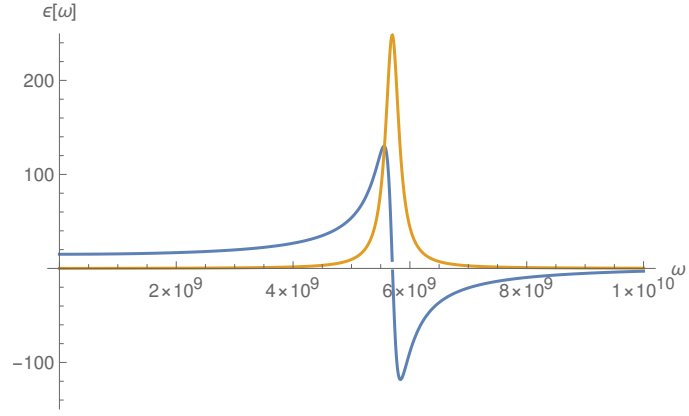


Figure 13: Imaginary (yellow) and real (blue) parts of the BST dielectric function

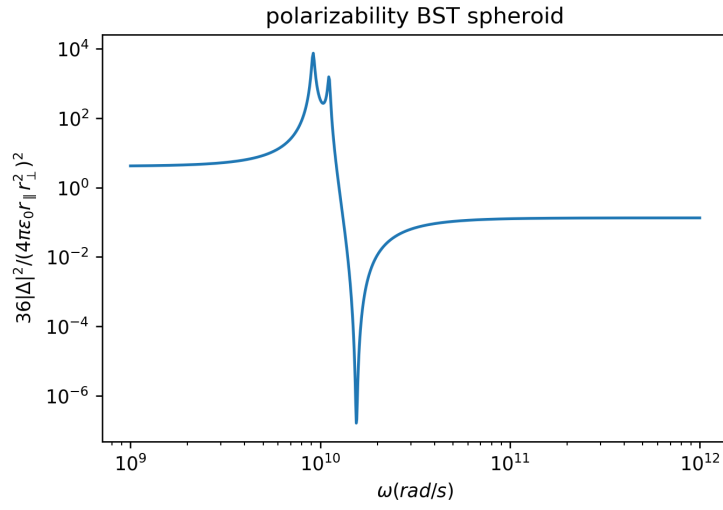


Figure 14: Modulus squared of the difference of the longitudinal and transverse polarizabilities.

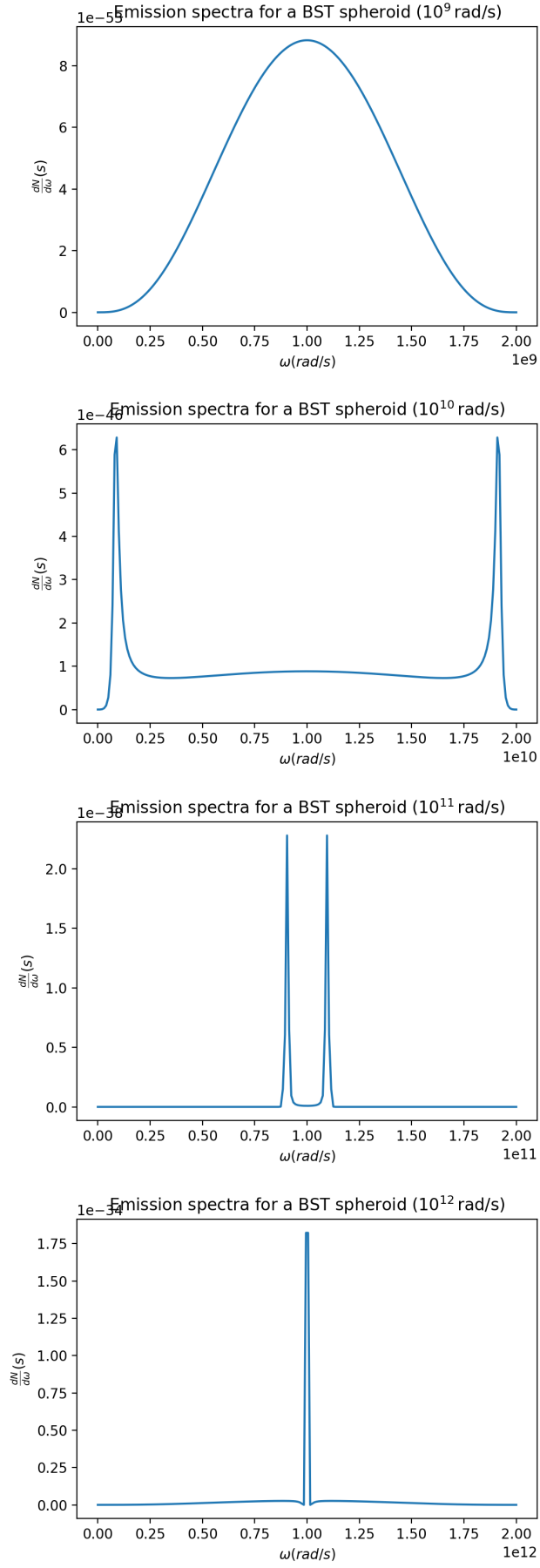


Figure 15: Emission spectra for a BST nanospheroid for different rotation velocities Ω (a) 10^9 rad/s (b) 10^{10} rad/s (c) 10^{11} rad/s (d) 10^{12} rad/s

6 Consistency tests

We present below a few consistency tests realized with an independent Python implementation. We present alternative plots based on Eq.(42).

We focus first on the estimation on the integral

$$I(\Omega, N_{\perp}, N_{\parallel}) = \int_{-1}^1 du (1-u^2)^3 \frac{1}{\left[\left(1 - \frac{\Omega^2 u^2}{\omega_p^2 N_{\perp}}\right)^2 + \left(\frac{\Omega \gamma u}{\omega_p^2 N_{\perp}}\right)^2 \right] \left[\left(1 - \frac{\Omega^2 u^2}{\omega_p^2 N_{\parallel}}\right)^2 + \left(\frac{\Omega \gamma u}{\omega_p^2 N_{\parallel}}\right)^2 \right]}. \quad (56)$$

and the corresponding emission rate is then given by a simple scaling factor as

$$\Gamma(\Omega, N_{\perp}, N_{\parallel}) = \frac{1}{9\pi} \frac{\Omega^7}{c^6} \frac{(r_{\parallel} r_{\perp}^2)^2}{36} \left(\frac{1}{N_{\parallel}} - \frac{1}{N_{\perp}} \right)^2 I(\Omega, N_{\perp}, N_{\parallel}) \quad (57)$$

We show the evaluation of these integrals with a simple Riemann sum below. As we use a “stupid” linear sampling with bins of identical dimensions on the interval $[0, \Omega]$, for $\Omega \gg \omega_p$ a larger number of points is needed. For $N_u = 10^5$, this simple integration yields satisfying results up to $\Omega \simeq 2 \times 10^{11}$ Hz = $20 \omega_p$. The maximum enhancement is, as expected, for Ω on the same order as the plasmon frequency ω_p (is it $2\omega_p$?).

The “angular point” in the emission rates occurs when the enhancement factor $I(\Omega, N_{\perp}, N_{\parallel})$ becomes much larger than one. If we decrease the sampling, “artefact peaks” in Ω appear as a result

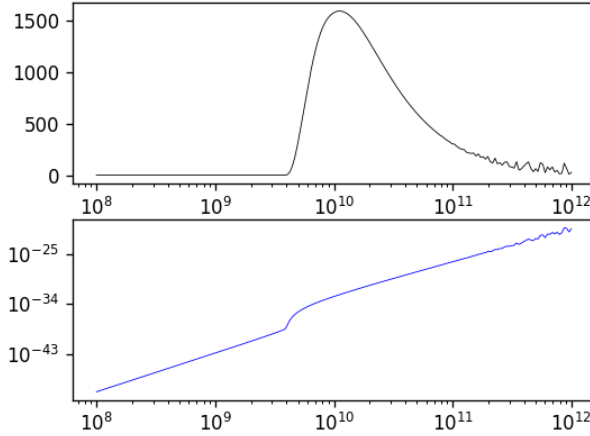


Figure 16: $I(\Omega, N_{\perp}, N_{\parallel})$ (upper figure) and $\Gamma(\Omega, N_{\perp}, N_{\parallel})$ (lower figure) vs Ω . $N_u = 10^5$ bins used for the Riemann sampling. Parameters used: $N_{\perp} = 0.15$, $N_{\parallel} = 0.70$, $r_{\perp} = 10^{-9}$ m, $r_{\parallel} = 2.10^{-9}$ m, $\gamma = 3.10^{-4}\omega_p$, $\omega_p = 10^{10}$ Hz, $r_{\perp} = 10^{-9}$ m, $r_{\parallel} = 2.10^{-9}$ m.

of an insufficient sampling in the numerical integration. See below, for instance, the integration with $N_u = 10^4$ (instead of $N_u = 10^5$).

We can also check the validity of the Lorentzian approximation

$$I_{\text{Lor}}(\Omega, N_{\perp}, N_{\parallel}) = 2 \int_{-\infty}^{\infty} du \left\{ \frac{(1-u_{\perp}^2)^3}{\left[4 \left(1 - \frac{u}{u_{\perp}}\right)^2 + \frac{\gamma^2}{\omega_p^2 N_{\perp}} \right] \left[\left(1 - \frac{u_{\perp}^2}{u_{\parallel}^2}\right)^2 + \frac{\gamma^2 N_{\perp}}{\omega_p^2 N_{\parallel}^2} \right]} + (u_{\parallel}, N_{\parallel} \leftrightarrow u_{\perp}, N_{\perp}) \right\} \quad (58)$$

With the additional factor 2, both integrals coincide with an excellent approximation in the high frequency regime $\Omega \geq \omega_p$.

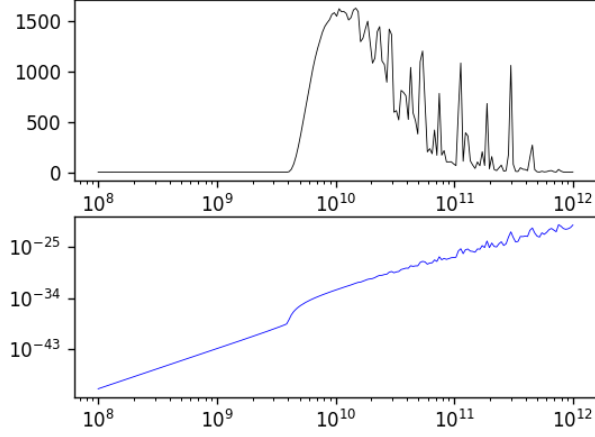


Figure 17: $I(\Omega, N_{\perp}, N_{\parallel})$ (upper figure) and $\Gamma(\Omega, N_{\perp}, N_{\parallel})$ (lower figure) vs Ω with an insufficient sampling - $N_u = 10^4$ bins - used for the Riemann sampling. Other parameters are identical to previous Figure.

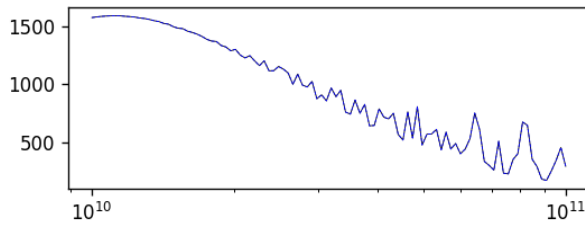


Figure 18: $I(\Omega, N_{\perp}, N_{\parallel})$ and $I_{\text{Lor}}(\Omega, N_{\perp}, N_{\parallel})$ vs Ω for frequencies s.t. $\omega_p \leq \Omega \leq 10 \omega_p$. $N_u = 10^4$ bins used for the sampling. Parameters identical to the previous Figure. Both curves are perfectly superimposed.